Chapter 5: Useful probability distributions

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1. Important discrete distributions

2. Some important continuous distributions

5.1 Important discrete distributions

Recall: The set of all possible values of a discrete variable together with their associated probabilities is called a **discrete probability distribution**.

Definition 1

A sequence of Bernoulli trials is a sequence of trials performed under conditions
(a) for each trial, there are only two possible outcomes, say, success and failure;
(b) the probabilities of the occurrence of these outcomes remain the same throughout the trials;
(c) the trials are carried out independently.

- Let denote event "success" by S, and event "failure" by F.
- Let $P(S) = p$, and $P(F) = q$, where $p + q = 1$.

Definition 2 (Binomial distribution)

Binomial distribution: The probability distribution of a random variable X representing the number of successes in a sequence of n Bernoulli trials

- $n =$ the number of trials
- $p =$ probability of success.
- X is said to have binomial probability distribution with parameters (n, p), denote $X \sim B(n, p)$
- $p_X(k) = \binom{n}{k}(1-p)^{n-x}p^k, k = 0, 1, \ldots, n,$
- $E(X) = np$
- $\mathrm{var}(X) = np(1-p)$

Example 3

A homeowner has just installed 20 light bulbs in a new home. Suppose that each has a probability 0.2 of functioning more than three months. What is the probability that at least five of these function more than three months? What is the average number of bulbs the homeowner has to replace in three months?

Solution. Let X be the number of bulbs functioning more than three months (success). X has a binomial distribution with $n = 20$ and $p = 0.3$. The answer to the first question is thus given by

$$P(X \geq 5) = \sum_{k=5}^{20} P(X = k) = 1 - \sum_{k=0}^{4} P(X = k)$$

$$= 1 - \sum_{k=0}^{4} \binom{20}{k} (0.2)^k (1 - 0.2)^{20-k} = \ldots = 0.37$$

The average number of replacements is

$$20 - E(X) = 20 - np = 20 - 20(0.2) = 16$$

R codes for $X \sim B(n, p)$

- Find $P(X = k)$.
  R code: **dbinom(k,n,p)**
- Find $P(X \leq k)$.
  R code: **pbinom(k,n,p)**

Example 4

Let $X \sim B(20, 0.2)$. Find $P(X \geq 5)$.
R code: 1 − pbinom(4,20,0.2) $\leftarrow (P(X \geq 5) = 1 - P(X \leq 4))$
Output: 0.3703517

Example 5

According to a survey of a country, 41% of all households are wireless-only households (no landline). In a random sample of 20 households, what is the probability that

a. exactly 5 are wireless-only?
b. fewer than 3 are wireless-only?
c. at least 3 are wireless-only?
d. the number of households that are wireless-only is between 5 and 7, inclusive?

Example 6

An exciting computer game is released. Sixty percent of players complete all the levels. Thirty percent of them will then buy an advanced version of the game.

a. Among 15 users, what is the expected number of people who will buy the advanced version?
b. What is the probability that at least two people will buy it?

Definition 7 (Geometric distribution)

For a sequence of indepndent Bernoulli trials, the number of trials needed to get the **first** success has **Geometric distribution**.

- $p =$ probability of success
- $X \sim Geo(p)$,
- Its PMF is

$$p(k) = \begin{cases} (1-p)^{k-1}p, & k = 1, 2 \ldots \\ 0, & \text{otherwise} \end{cases}$$

- $E(X) = \dfrac{1}{p}$ and $\mathrm{var}(X) = \dfrac{1-p}{p^2}$

R code

a. Find $P(X = k)$. R code: **dgeom(k-1, p)**

b. Find $P(X \leq k)$. R code: **pgeom(k-1,p)**

Example 8

Roll a die repeatedly until you observe face 1 for the first time.
Let Y be the total number of die rolls.
Then Y is a geometric random variable with parameter $p = \frac{1}{6}$.

Example 9

A certain basketball player in NBA makes 70% of his free throws. What is the probability that he

a. misses his first two free throws and makes the third one.

b. makes his first or second free throws.

c. makes his first free throw after four attempts.

Solution. Let X be the number of free throws when the first success occurs. X has geometric probability distribution with $p = 0.7$.
a. $P(X = 3) = 0.063$
b. $P(X \leq 2) = 0.91$
c. $P(X \geq 5) = 0.0081$

Definition 10 (Negative Binomial distribution)

In a sequence of independent Bernoulli trials, the number of trials needed to obtain k successes has **Negative Binomial distribution**.

- k : number of success
- p : probability of success
- X has a negative binomial distribution with k and p : $X \sim NB(k, p)$
- $p_X(x) = \binom{x-1}{k-1}(1-p)^{x-k}p^k$
- $E(X) = \dfrac{k}{p}$ and $\mathrm{var}(X) = \dfrac{k(1-p)}{p^2}$

R code $X \sim NB(k, p)$

a. Find $P(X = x)$. R code: **dnbinom(x-k, k, p)**

b. Find $P(X \leq x)$. R code: **pnbinom(x-k, k, p)**

Example 11

In a recent production, 20% of certain electronic components are defective. We need to find 3 non-defective components for our 3 new computers. Components are tested until 3 non-defective ones are found. What is the probability that 7 components will have to be tested?

Solution. Let X be the number of components which are tested until 3 non-defective ones are found. Then $X \sim NB(3, 0.2)$. The probability that 7 components will have to be tested

$$P(X = 7) = \binom{7-1}{3-1}(0.8)^4(0.2)^3 = 0.049152$$

R code: dnbinom(4, 3, 0.2)

Assume that the numbers of occurrences in disjoint intervals are independent and the mean of the numbers of occurrences is constant.

Example 12

1. The number of cars passing through a traffic light **in 1 hour.**
2. The number of requests to web servers **in an hour**.
3. Number of phone calls **per minute.**
4. The number of mistakes in a book **per page**.

Definition 13 (Poisson distribution)

A discrete random variable X is said to follow the Poisson probability distribution with parameter $\lambda > 0$, denoted by $P(\lambda)$, if

$$P(X = x) = e^{-\lambda}\frac{\lambda^x}{x!},\ x = 0, 1, 2, \ldots$$

- λ : average number of events
- Denote $X \sim P(\lambda)$

Table A.2 in Appendix A gives mass function

- Siméon Denis Poisson (1781 - 1840)
- A French mathematician and physicist
- Source: https://en.wikipedia.org

Example 14

Let X be a Poisson random variable with $\lambda = \frac{1}{2}$. Find

1. $P(X = 0)$.
2. $P(X \geq 3)$.

Solution.

a. $P(X = 0) = e^{-1/2}\frac{(1/2)^0}{0!} = e^{-1/2} = 0.60653$

b. $P(X \geq 3) = 1 - P(X \leq 2) = 1 - (P(X = 0) + P(X = 1) + P(X = 2)) = \ldots = 0.01439$

Example 15

Customers of an internet service provider initiate new accounts at the average rate of 10 accounts per day.

a. What is the probability that more than 8 new accounts will be initiated today?

b. What is the probability that more than 16 accounts will be initiated within 2 days?

Solution. .....

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Approximate $\mathrm{B}(n,p) \approx \mathrm{P}(\lambda)$

Poisson distribution can be used to approximate Binomial distribution when the number of trials n is large, and the probability of success p is small.

$$\mathrm{B}(n,p) \approx \mathrm{P}(\lambda),$$

where $n \geq 30, p \leq 0.05$ and $np = \lambda$

Example 16

Three percent of electronic messages are transmitted with error. What is the probability that out of 200 messages, at least 5 will be transmitted incorrectly?

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5.2 Some important continuous distributions

Definition 17 (Uniform distribution)

A continuous random variable X has a **uniform distribution** over an interval a to b $(b > a)$ if it is equally likely to take on any value in this interval.

- Denote $X \sim U(a, b)$
- The probability density function (pdf) of X is constant over (a, b)

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

- $E(X) = \dfrac{a+b}{2}$ and $\text{var}(X) = \dfrac{(b-a)^2}{12}$

R command: $X \sim U(a, b)$

1. Find $P(X \leq k)$. **R code:** punif(k, a, b)

Example 18

If X is a uniformly distributed random variable over $(0, 10)$, calculate the probability that (a) $X < 3$, (b) $X > 6$, and (c) $3 < X < 8$.

Solution.

a. $P(X < 3) = \int_0^3 f(x)\mathrm{d}x = \int_0^3 \frac{1}{10-0}\mathrm{d}x = \frac{3}{10}$

b. $P(X > 6) = \int_6^{10} f(x)\mathrm{d}x = \int_6^{10} \frac{1}{10-0}\mathrm{d}x = \frac{4}{10}$

c. $P(3 < X < 8) = \int_3^8 f(x)\mathrm{d}x = \int_3^8 \frac{1}{10-0}\mathrm{d}x = \frac{5}{10}$

Example 19

A bus arrives at a bus stop every 8 minutes. What is the chance that the bus will arrive in 5 minutes or less if you arrive at the bus stop?

Solution. let X be the number of minutes that the bus will arrive in. It then has a uniform distribution $U(0, 8)$ (the minimum time is 0 minutes and the maximum time is 8 minutes) with pdf

$$f(x) = \begin{cases} \frac{1}{8}, & 0 \leq x \leq 8 \\ 0, & \text{elsewhere} \end{cases}$$

The probability that the bus will arrive in 5 minutes or less

$$P(X \leq 5) = \int_0^5 \frac{1}{8} \mathrm{d}x = \frac{5}{8}$$

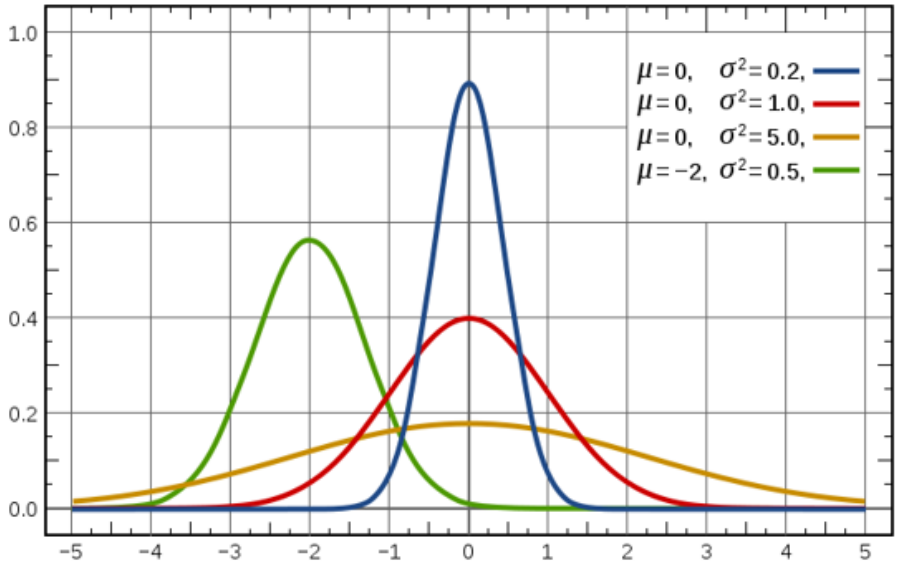
R code: punif(5, 0, 8)

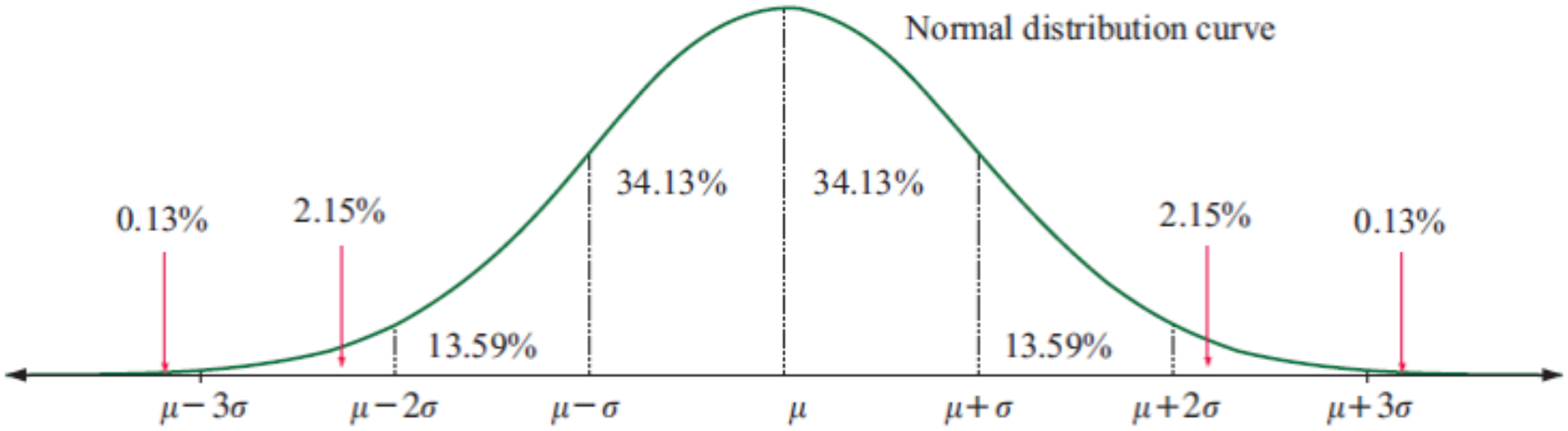
Definition 20

A random variable X is said to have a **normal probability distribution** with parameters μ and σ^2, if it has a pdf given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma}}$$

- Denote $X \sim N(\mu, \sigma)$
- $E(X) = \mu$ and $\mathrm{var}(X) = \sigma^2$





Notice that:

- $\approx 68.26\%$ of values lie between $\mu - \sigma$ and $\mu + \sigma$
- $\approx 95.44\%$ of values lie between $\mu - 2\sigma$ and $\mu + 2\sigma$
- $\approx 99.74\%$ of values lie between $\mu - 3\sigma$ and $\mu + 3\sigma$.

Standard normal probability distribution

- If $\mu = 0$ and $\sigma = 1$, we call X a standard normal random variable, denoted

$$X \sim N(0, 1)$$

- If $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X - \mu}{\sigma} \sim N(0, 1)$
- Let $X \sim N(0, 1)$, then $P(X \leq x)$ is given in Table A4

R command: $X \sim N(\mu, \sigma^2)$

1. Find $P(X \leq a)$. **R code:** pnorm(a, μ, σ)
2. Find a such that $P(X \leq a) = t$. **R code:** qnorm(t, μ, σ)

Example 21

a. Let $X \sim N(0,1)$. Find $P(X \geq 1.13)$

b. Let $X \sim N(0,1)$. Find t such that $P(X \geq t) = 0.0504$

c. Let $X \sim N(5,4)$. Find $P(-2.5 < X < 10)$

Solution. a. Using the normal table A4,

$$P(X \geq 1.13) = 1 - P(X < 1.13) = 1 - 0.8708 = 0.1292$$

Table A4, continued. Standard Normal distribution

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830

b. Using the normal table A4, $t = 0.56$

Table A4, continued. Standard Normal distribution

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
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0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
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0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767

c. Let $Z = \dfrac{X-\mu}{\sigma} = \dfrac{X-5}{2}$. Then $Z \sim N(0,1)$ and using the table A4

$$\begin{aligned} P(-2.5 < X < 10) &= P\left(\frac{-2.5-5}{2} < \frac{X-5}{2} < \frac{10-5}{2}\right) \\ &= P(-3.75 < Z < 2.5) \\ &= P(Z < 2.5) - P(Z < -3.75) \\ &= 0.9938 - 0.0001 = 0.9937 \end{aligned}$$

R code:

a) 1−pnorm(1.13, 0, 1)

b) qnorm(0.004, 0, 1)

Example 22

The scores of an examination are assumed to be normally distributed with $\mu = 75$ and $\sigma^2 = 64$. What is the probability that a student score chosen at random will be greater than 85?

Solution. Let X be a score of the exam. Then, $X \sim N(75, 64)$ and

$$\begin{aligned} P(X > 85) &= P\left(\frac{X-75}{8} > \frac{85-75}{8}\right) \\ &= P(Z > 1.25) = 0.1056 \end{aligned}$$

Definition 23

The gamma function, denoted by $\Gamma(a)$, is defined as:

$$\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} \mathrm{d}x.$$

- $\Gamma(a) = (a-1)\Gamma(a-1)$ for all $a > 1$.
- If $n \in \mathbb{N}^*$, then $\Gamma(n) = (n-1)!$

Student t-distribution

Definition 24

Student's t-distribution with degree of freedom n has the probability density function (PDF) given by

$$f(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}\left(1+\frac{t^2}{n}\right)^{-\frac{n+1}{2}}.$$

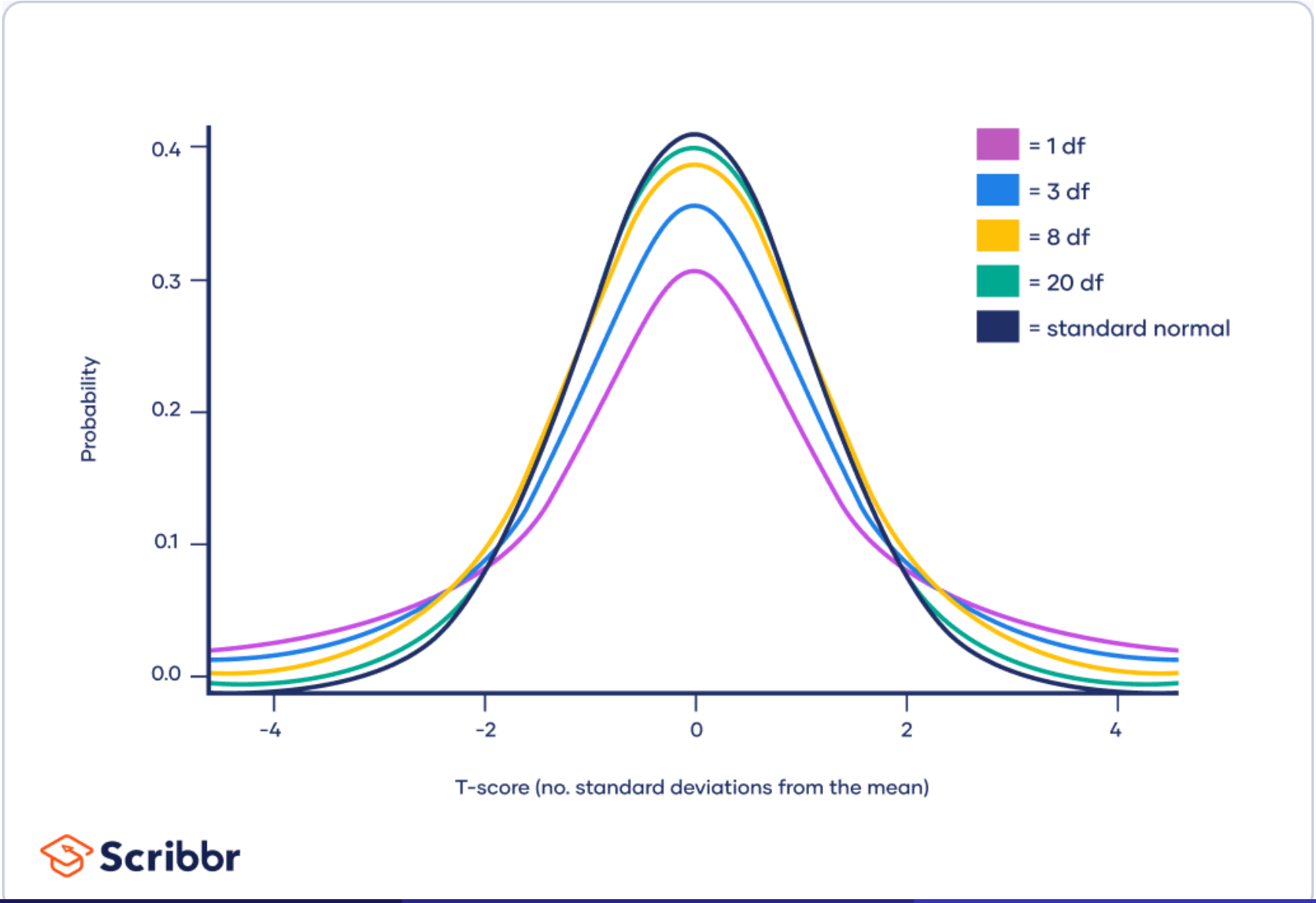
The t-distribution tends to a standard normal distribution as the degrees of freedom (equivalently, the sample size n) tend to infinity.

In t-table, we can find t_α, ν such that

$$P(T > t_{\alpha,\nu}) = \alpha.$$

ν (d.f.)	α, the right-tail probability								
	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	3.078	6.314	12.706	15.89	31.82	63.66	127.3	318.3	636.6
2	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.894	6.869
6	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587

The standard normal distribution provides a good approximation to the t-distribution for sample sizes of 30 or more.



Example 25

Let T be a random variable. Assume that T has t-distribution with 9 degree of freedom and $\alpha = 0,01$.

It follows from t-table, we have

$$P(T > 2,821) = 0,01.$$

R command. Let $X \sim T(n)$

- Find $P(X \leq x)$

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pt(x,n)
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References

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